Hints: Abstract algebra – 3. Permutations

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For $n = 4$: the Klein four-group $V = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\}$ is normal in S_4 and $S_4/V \cong S_3$. Take $\pi_1 = (12)(34)$. The image of $\langle \pi_1, \pi_2 \rangle$ in S_3 is cyclic, so it is not all of S_3.

Hint 2. For $n = 3$ and $n = 5$ it suffices to treat one representative of each cycle type, since conjugating π_1 and π_2 together preserves the property. Useful tools: a transposition and an n-cycle on adjacent points generate S_n (with n prime even any transposition works), and an even and an odd permutation cannot generate a subgroup of A_n.

Hint 3. Partners in S_5 that work: (12) with (2345); (123) with (2345); (1234) with (45); (12345) with (45); (12)(34) with (2354); (123)(45) with (34). To check each one, find a transposition and a 5-cycle in the generated group, or count orders. In S_3, pair a transposition with a 3-cycle.

Problem 2

Hint 1. Make a graph on $\{1, \ldots, n\}$ with an edge $\{i, j\}$ for each generating transposition (ij). Show that $S(k)$ is exactly the connected component of k.

Hint 2. A graph with n vertices and $n - 2$ edges is not connected (a connected graph needs at least $n - 1$ edges), so every component has at most $n - 1$ vertices.

Problem 3

Hint 1. Count fixed points over the whole group. Let $m = |G| > 1$. Every non-identity element fixes exactly one point, so the stabilisers G_x intersect pairwise trivially, and the non-identity elements are partitioned according to their fixed point.

Hint 2. For an orbit O with stabiliser order t_O, the number of non-identity elements fixing some point of O is $|O|(t_O - 1) = m - m/t_O$. Summing over orbits gives $m - 1 = \sum_O m(1 - 1/t_O)$.

Hint 3. Each orbit with $t_O \geqslant 2$ contributes at least $m/2$, so exactly one orbit has $t_O > 1$. Then $m - 1 = m - m/t_O$ gives $t_O = m$: that orbit is a single point fixed by all of G.

Problem 4

Hint 1. Use the generating function $\sum_\pi x^{\mathrm{inv}(\pi)} = \prod_{k=1}^{n}(1 + x + \cdots + x^{k-1})$ and the roots-of-unity filter with $\zeta = e^{2\pi i/p}$, $n = p - 1$.

Hint 2. For $p \nmid j$, $F(\zeta^j) = \prod_{k=1}^{p-1} \frac{1-\zeta^{jk}}{1-\zeta^j} = \frac{p}{(1-\zeta^j)^{p-1}}$. Hence $f(p-1) - \frac{(p-1)!}{p} = \sum_{j=1}^{p-1}(1-\zeta^j)^{-(p-1)}$.

Hint 3. Write $1 - \zeta^j = -2i \sin(\pi j/p) e^{i\pi j/p}$. The terms $j = 1$ and $j = p - 1$ dominate, and their sum is $(-1)^{(p+1)/2}\, 2\cos(\pi/p)/(2\sin(\pi/p))^{p-1}$. The others are smaller by a factor of roughly $(2\cos(\pi/p))^{-(p-1)}$.

Hint 4. So the difference is positive for $p \equiv 3 \pmod 4$ and negative for $p \equiv 1 \pmod 4$ (check: $p = 7$ gives $107 > 102.86$, $p = 5$ gives $4 < 4.8$). There are infinitely many primes in each class.

Problem 5

Hint 1. An n-cycle has sign $(-1)^{n-1}$. For odd n, two n-cycles generate a subgroup of A_n, which is not S_n (except for the degenerate case $n = 1$).

Hint 2. For even n (for $n = 2$ it is trivial) take $c_1 = (1\,2\,\ldots\,n)$ and $c_2 = c_1(1\,2\,3)$. Check that $c_2 = (1\,3\,2\,4\,5\ldots n)$ is again an n-cycle.

Hint 3. The group $\langle c_1, c_2\rangle$ contains (123) and its conjugates by powers of c_1, i.e. all $(k, k{+}1, k{+}2)$, which generate A_n. Since c_1 is odd, the group is S_n. Answer: all even n (and $n = 1$ trivially).

Problem 6

Hint 1. n pairwise different cyclic shifts are all the cyclic shifts. Up to a row permutation (a sign), the matrix is the circulant C with first row $a_1, \ldots, a_n$.

Hint 2. $\det C = \prod_{j=0}^{n-1} f(\omega^j)$, where $f(z) = \sum_k a_k z^{k-1}$ and $\omega = e^{2\pi i/n}$ (the vectors $(1, \omega^j, \omega^{2j}, \ldots)$ are eigenvectors). For $j = 0$ the factor is $\sum a_k = \frac{n}{2}(a_1 + a_n)$.

Hint 3. For $z = \omega^j \neq 1$ and $a_k = a_1 + (k-1)d$: $f(z) = d\sum_{m=0}^{n-1} m z^m = \frac{dn}{z-1} \neq 0$ because $d \neq 0$.

Problem 7

Hint 1. Think of $\pi(1), \ldots, \pi(n)$ as a walk visiting every value once with steps of size at most 2. Let a_n be the number of such walks starting at the value 1. Show $a_n = a_{n-1} + a_{n-3} + 1$ (after 1 comes 2; or 3, 2, 4; or the walk $1, 3, 5, \ldots$ up the odd numbers and back down the even ones).

Hint 2. If 1 is interior, its neighbours are 2 and 3, and from 1 the two branches are forced to go $2, 4, 6, \ldots$ and $3, 5, 7, \ldots$ until one of them stops at some value $m \in \{2, \ldots, n-1\}$. After that, the rest is a walk on $\{m{+}1, \ldots, n\}$ starting at $m{+}1$. Deduce, for $n \geqslant 2$, $P_n = 2\big(a_n{+}a_1{+}a_2{+}\cdots{+}a_{n-2}\big)$.

Hint 3. Plug this in and use the recurrence for a_n twice. The quantity equals 4 for all $n \geqslant 2$ (e.g. $P = 1, 2, 6, 12, 20, 34, 56, 88, \ldots$).

Problem 8

Hint 1. Look for a choice that makes $x_k x_{k+1}$ telescope modulo p, like $\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}$.

Hint 2. Take $x_k \equiv k^{-1} \pmod p$, which is a permutation of $1, \ldots, p-1$. Then the sum is $\sum_{k=1}^{p-2}(\frac{1}{k} - \frac{1}{k+1}) = 1 - \frac{1}{p-1} \equiv 1 - (-1) = 2$.

Problem 9

Hint 1. Following the hint, $\det(x^{|i-j|}) = \sum_\sigma \operatorname{sgn}(\sigma) x^{D(\sigma)}$. Compute this determinant in closed form.

Hint 2. Subtract x times row i from row $i+1$ (going from the bottom up). Below the diagonal everything vanishes, and you get $\det = (1 - x^2)^{n-1}$, a polynomial of degree $2n - 2$.

Hint 3. Reduce modulo 2, where signs disappear: $\sum_d Q(n, d)x^d \equiv (1 - x^2)^{n-1} \pmod 2$. So $Q(n, d)$ is even for every $d > 2n - 2$.

Problem 10

Hint 1. A term $\prod_i m_{i\sigma(i)}$ of $\operatorname{per}(J - I)$ is 1 exactly when σ has no fixed points, and 0 otherwise.

Hint 2. Write each row of $J - I$ as (row of ones) $- e_i$ and expand multilinearly, as one does for determinants. Choosing e_i in the rows of a set S forces σ to fix S, and leaves $\operatorname{per} = (n - |S|)!$. So $D_n = \sum_k (-1)^k \binom{n}{k}(n-k)!$.

Hint 3. If $\sigma(i) = j$ and $\sigma(l) = k$, let τ be σ with these two values swapped. Then $P_\sigma - P_\tau = E_{ij} + E_{lk} - E_{ik} - E_{lj}$. These "rectangles" span the matrices with all line sums 0, of dimension $(n-1)^2$; adding I gives every matrix with equal row and column sums.